

Final Exam AI Study Tips

1 Heuristic Search

- Understand what is heuristic?
- Understand A* and Best-First Search.
- Learn heuristic properties: monotonicity/consistency, admissibility, informedness
- Prove that the solution path or $h(n)$ is admissible and consistent.
- Practice solving pathfinding problems.

2 Heuristic in Games

- Understand the concept of Minimax algorithm for decision-making.
- Understand the concept of alpha beta pruning algorithm for decision-making.
- What is the role of alpha and beta in alpha beta pruning algorithm.
- What is the rule of pruning in alpha beta pruning.
- Learn Alpha-Beta Pruning to optimize game search.
- Apply these concepts to simple board games/searching graph.

3 Production System

- Know components: rules, working memory, control strategy.
- Differentiate between **data-driven (forward chaining)** and **goal-driven (backward chaining)** approaches.
- Practice and apply these concepts to simple real world problem.
-

4 Intelligent Agent

- Understand what is agent? intelligent agent and rational agent.
- Learn **PEAS (Performance, Environment, Actuators, Sensors)** framework and PAGE .
- Compare agent types: Reflex, Model-Based, Goal-Based, Utility-Based, Learning Agents and etc.
- Identify real-world examples (e.g., self-driving cars, recommendation systems with PEAS/AGENT design architecture)

5 Machine Learning (ML)

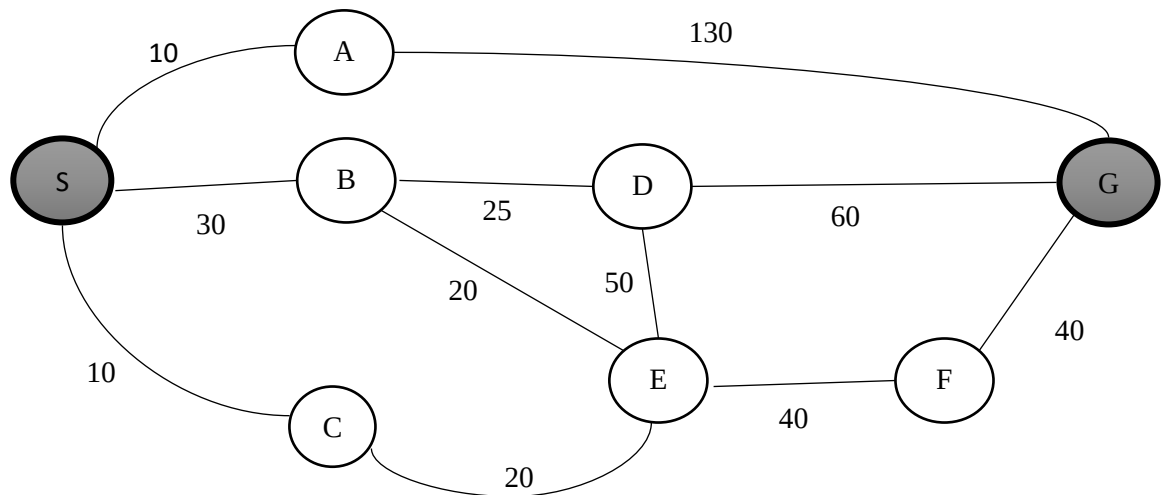
- Differentiate **supervised vs. unsupervised learning and reinforcement**.
- Understand classification, clustering, regression. What are the differences among them?
- Common type of Classification, clustering and classification model
- Learn **confusion matrix** (True Positive, False Negative, etc.).
- Recognize **high-risk vs. low-risk errors** in ML models.
- Study the situation in **High recall, low precision and low recall, high precision**.
- Apply recall, precision and accuracy in real problem.

🔥 Understand the concepts deeply, practice past year questions to familiarize yourself with the exam format, review and complete class exercises along with their solutions, apply what you learn through problem-solving and stay consistent in your studies to achieve the best results! 🚀

EXAMPLE PAST YEAR QUESTIONS

QUESTION 1

Below is a symmetric undirected graph of the airline route within 8 nodes (n) from airport city S to city G. The actual distance is denoted in kilometres (km) from one node to the other node. For example, actual distance from airport city S to A is 10 km and airport city S to B is 30 km.



Given the values of $S = 50$, $A = 40$, $B = 60$, $C = 30$, $D = 70$, $E = 50$, $F = 50$, $G = 0$

which are the straight path distance from node (n) to G:

- Find the total minimal travel distance from airport city, S to G using A* algorithm and evaluate whether the solution path from S to G is admissible. Your answer should in tabular form with with following columns: $h(n)$ and $h^*(n)$;
- Evaluate the solution path obtained in (a) is consistent. Your answer should in tabular form with following columns : $h(n)$, $c(n,n')$, $h(n')$, and $c(n,n') + h(n')$.

Question 2 (A)– Alpha beta Pruning

Apply alpha-beta search (from left to right) to the following game tree as shown in Figure 3 below.

- Show the backed-up value of each node (no values for nodes that are not examined).
- Indicate all parts of the tree that are pruned / cutoffs by marking the line with double slash (**X**) indicator.
- Calculate the α and β final values for node B? (2 marks)
- Calculate the optimal value for MAX to win the game?

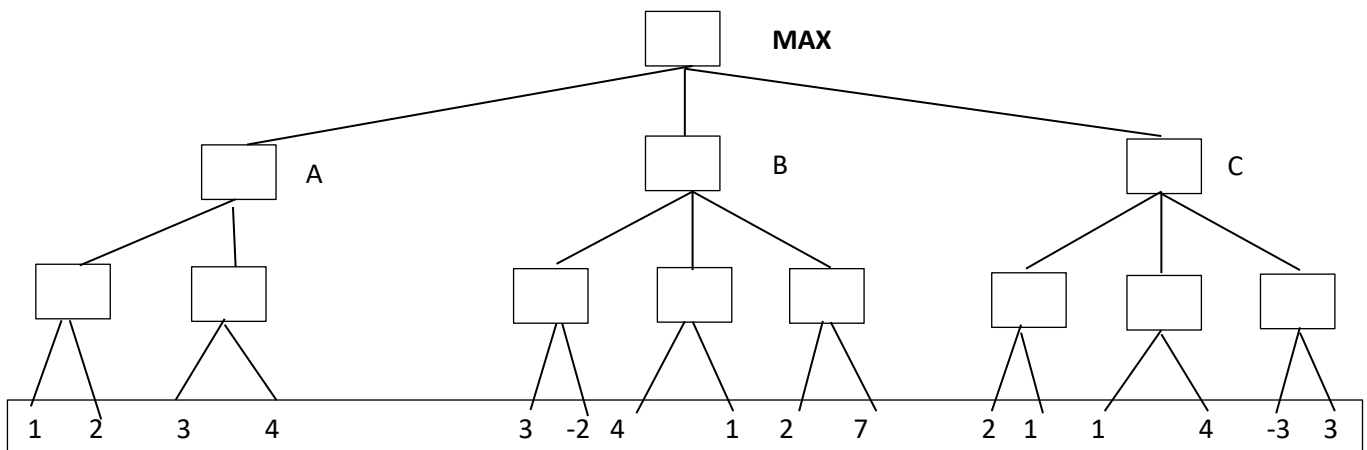
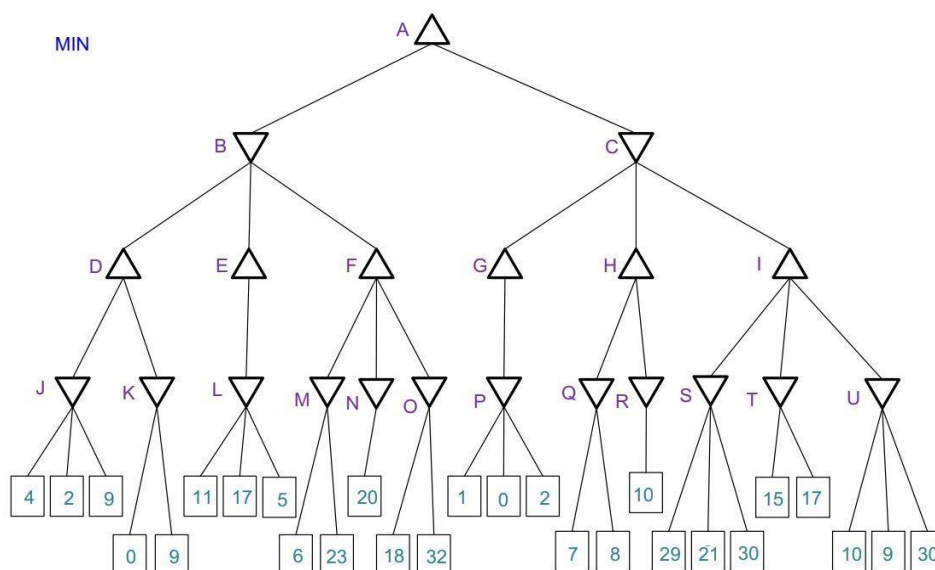


Figure 3

Q2B. Complete all states (A – U) in the following tree using MINIMAX without Alpha Beta pruning. (10 m)



*** Using drop down / multi-choice to select from A - U

Question 3 – Production Systems

Consider the following knowledge in a production system intended to describe the reasoning of a healthy lifestyle. The information is as follows:

EatJunkFood	:	whether or not a person eats junk food every now and then
Exercise	:	whether or not a person exercises
HealthyDiet	:	whether or not a person has a healthy diet
SleepRight	:	whether or not a person has a healthy sleep schedule

The facts and rules are listed below:

Fact 0: EatJunkFood

Fact 1: Exercise

Fact 2: HealthyDiet

Fact 3: SleepRight

Rule 1: $\text{Exercise} \wedge \text{HealthyDiet} \rightarrow \text{LoseWeight}$

Rule 2: $\text{SleepRight} \wedge \text{HealthyDiet} \rightarrow \text{FeelGood}$

Rule 3: $\text{FeelGood} \wedge \text{EatJunkFood} \wedge \text{LoseWeight} \rightarrow \text{MaintainWeight}$

- a. Use **FORWARD CHAINING** to show a production system trace of the sentence MaintainWeight (5 marks)

Use **BACKWARD CHAINING** to show a production system trace of the sentence MaintainWeight (5 marks)

QUESTION 4

Let,

A = the ignition key is on

B = the engine will not start

C = headlights work

D = the voltage test on the ignition switch shows 1 to 6 volts

(A, B, C and D are the FACTS in the Initial Database)

E = the starting system (including battery) is faulty

G = starter is faulty

I = battery is dead

F = wiring between the ignition and the solenoid is good

H = replace the ignition switch

(E, F, G, H, I are the FACTS that can be inferred from the FACTS in the Initial Database)

Rules are given below and they are in the order of R1, R2, R3, R4 and R5:

R1: $A \wedge B \rightarrow E$

R2: $E \wedge C \rightarrow G$

R3: $E \wedge \sim C \rightarrow I$

R4: $D \rightarrow F$

R5: $F \rightarrow H$

Consider conflict set as a queue and rule must be fired from front of the queue

(a) Use the forward chaining to infer all the possible FACTS from Initial Database (10M)

Iteration #	Working Memory	Conflict Set	Rule Fired
1	A, B, C, D		
2			
3			
4			
5			

QUESTION 5 MACHINE LEARNING

Q6. Figure A shows Neighbourhood X attributes in urban areas and their binary classification results. Below are the detailed attributes.

Household	Gender	Car_type	Property	Class_T20	Predicted_classT20
1	M	Family	Semi_d	T20	T20
2	M	Sports	Bungalow	M40	M40
3	M	Sports	Semi_d	M40	M40
4	M	Sports	Semi_d	T20	T20
5	M	Sports	Semi_d	M40	T20
6	M	Luxury	Semi_d	M40	T20
7	F	Luxury	Semi_d	M40	M40
8	F	Luxury	Terrace	T20	T20
9	F	Luxury	Bungalow	M40	M40
10	F	Luxury	Terrace	M40	M40
11	M	Luxury	Semi_d	M40	M40
12	M	Family	Bungalow	T20	T20
13	M	Family	Bungalow	T20	T20
14	M	Sports	Bungalow	M40	M40
15	F	Family	Terrace	T20	T20
16	F	Sports	Semi_d	T20	M40
17	F	Sports	Terrace	T20	M40
18	F	Family	Terrace	T20	M40
19	F	Family	Terrace	T20	T20
20	F	Luxury	Semi_d	M40	M40

Based on the dataset given in Figure A, answer the following questions:

a) Quantify the amount of True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN) for the classification result.

b) Using your answers in a), calculate the accuracy, precision, and recall for this classification